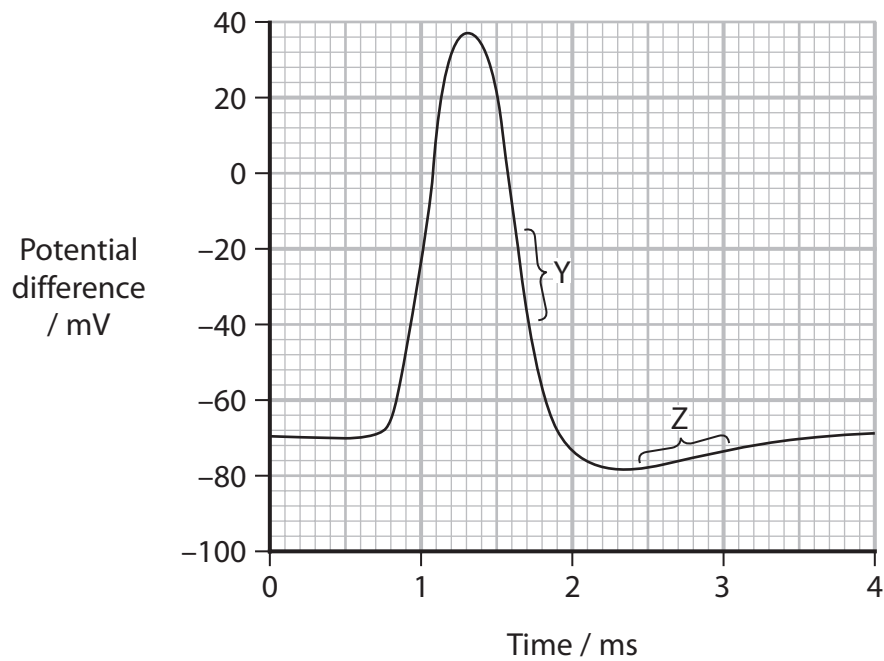


1 Nerve impulses travel along axons as action potentials.

(a) The graph shows an oscilloscope trace for an action potential in an axon.



(i) Which is the resting potential for this axon?

(1)

- ☐ **A** -70 mV
- ☐ **B** -78 mV
- ☐ **C** 38 mV
- ☐ **D** 108 mV

- (ii) Both voltage-gated sodium and voltage-gated potassium ion channels are involved in generating an action potential.

Which of these voltage-gated channels are open at Y?

(1)

- ☐ **A** no voltage-gated ion channels are open
- ☐ **B** voltage-gated sodium ion channels only
- ☐ **C** voltage-gated potassium ion channels only
- ☐ **D** both sodium and potassium voltage-gated ion channels

- (iii) Which is the state of polarisation of the membrane at Z?

(1)

- ☐ **A** depolarised
- ☐ **B** hyperpolarised
- ☐ **C** hypopolarised
- ☐ **D** unpolarised

- (iv) How many of the following statements are correct?

- the magnitude of the action potential is proportional to the strength of stimulus that generates the action potential
- action potentials spread out in both directions along the axon
- following an action potential, there is a refractory period during which it is not possible to generate a new action potential

(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three

(b) Explain why maintaining a resting potential requires ATP.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

**(Total for Question 1 = 7 marks)**

---